Predicting Algal Blooms with High Resolution Imaging and Machine Learning

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Abstract

The ongoing rise in ocean sea temperatures due to influencing factors of Climate Change has caused different anomalies to our oceans environmental conditions. These alterations have contributed to the increased prevalence and intensity of harmful algal blooms (HABS) specifically, the Lingulodinium polyedra. This species contributes to red tide events and bioluminescent activity along the California coast while producing potentially harmful yessotoxins associated with marine organism mortality and negative human health impacts. Respiratory complications and cutaneous irritation (dermatitis) are just a few of the various symptoms experienced by individuals exposed to harmful algal blooms. Due to this, studies have increased to understand the variability, response, and underlying mechanisms of harmful algal blooms in relation to climate change. Additionally due to an uptick of public health concerns, it is vital to understand major bloom events to further assess the risk and protocol to mitigate future outbreaks. This study aimed to develop a predictive framework for harmful algal blooms associated with L. polyedra by using imaging flow cytobot (IFCB) data and environmental variables, applying optimized machine learning models and creating an interactive data visualization platform to analyze bloom patterns and temporal dynamics. We used multiple modeling approaches to estimate the abundance of L. polyedra during harmful algal bloom events, including Linear Regression, Ridge Regression, Random Forest, univariate Simplex projection, and Multiview Embedding. This allowed us to compare standard machine learning baselines with time aware forecasting methods and identify which approach performed best across different prediction horizons. By combining these model results with interactive visualizations of bloom dynamics, this project aims to improve public understanding, support environmental monitoring, and help reduce the ecological and community impacts of harmful algal blooms.

Website: https://github.com/lemoncastle/hires-algal-bloom-prediction
Code: https://lemoncastle.github.io/hires-algal-bloom-prediction/

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1 Introduction

1.1 Overview

This report focuses on classified images of the dinoflagellate *Lingulodinium polyedra*, a species that is potentially harmful being producers of a group of toxins known as yessotoxins (Paz et al. 2004), with blooming events happening all around the world. For this study we aim to predict harmful algal blooms associated with Lingulodinium polyedra, featuring mostly from the California coast. We are using Imaging flow cytobot (IFCB) data along with environmental variables to achieve our primary objective. Baseline regression models are first implemented and compared with optimized algorithms to improve predictive performance.In addition to model development, an interactive data visualization platform is designed to display spatial and temporal bloom patterns. We hope that with the combination of both, our stakeholders are able to promote, use, and identify nuances in our findings to further the research of HABs as a whole. (Mertens et al. 2023)

1.2 Motivation

Lingulodinium polyedra is a single-celled, bioluminescent dinoflagellate that forms recurrent red tides along the Southern California coast, with documented occurrences dating back to at least the early 1900s (Skelton et al. 2024). Although these blooms produce visually striking nighttime bioluminescence, they represent a serious environmental and public health concern. *L. polyedra* is classified within Harmful Algal Blooms (HABs) because it produces yessotoxins (YTXs), polycyclic ether compounds that are recognized as potent cytotoxins and for which the European Union has established maximum permitted levels in shellfish (Mertens et al. 2023). Beyond direct toxicity, dense blooms drive severe hypoxic and hyperoxic conditions in the water column as organic matter accumulates and decays, creating conditions lethal to fish and benthic invertebrates (Skelton et al. 2024).

The scale and consequences of these events have become increasingly clear. During the spring of 2020, an exceptional bloom of *L. polyedra* extended over 400 km from northern Baja California to the Santa Barbara Channel, reaching the highest cell densities recorded in over a century of monitoring at Scripps Institution of Oceanography (Skelton et al. 2024). The bloom persisted for more than six weeks and was followed by mass die-offs of fish, crabs, mussels, and invertebrates along San Diego, Orange County, and northern Baja California coastlines, as well as significant losses at aquaculture facilities and research aquaria (National Centers for Coastal Ocean Science 2020). A survey of beachgoers during this event found that 25% reported symptoms including sore throat, watering eyes, and dermatitis (Wilson et al. 2022). A parallel study by Ternon et al. (2023) provided the first direct measurements of YTX in coastal aerosols associated with an *L. polyedra* bloom, demonstrating that yessotoxin and homo-yessotoxin were detectable in onshore sea spray during the 2020 event, with aerosol concentrations correlated with westerly winds and higher tides. These findings raise important questions about airborne exposure pathways for both marine organisms and coastal communities.

Prior research has also documented lethal effects on marine wildlife. Wit et al. (2014) examined coastal California sites before and after bloom conditions and found strong forensic genomic evidence linking algal toxins to abalone die-offs, with yessotoxins detected in digestive tissues. More broadly, Anderson et al. (2021) document that the diversity of HAB species and toxins affecting U.S. coastal waters has grown substantially over recent decades, with more regions, resources, and time intervals now requiring active monitoring. The economic toll is significant: HABs are estimated to have cost U.S. commercial fisheries and aquaculture over $5.6 billion in 2018 alone (Anderson et al. 2021). Wider ecological concerns are further supported by Hallegraeff et al. (2021), whose analysis of 33 years of global bloom records found that while uniform global trends are difficult to establish, regional bloom dynamics and emergent impacts are highly variable, underscoring the need for localized, high-resolution monitoring.

Despite these risks, HAB monitoring along the Southern California coast is often delayed, fragmented, or difficult for the public to interpret in real time (Anderson et al. 2021). The California Harmful Algal Bloom Monitoring and Alert Program (HABMAP), while valuable, relies primarily on weekly pier sampling at a limited number of fixed stations, which can miss rapidly evolving bloom dynamics between sampling intervals (Clarissa Anderson, Megan Hepner-Medina 2020). This project addresses that gap by combining Imaging FlowCytobot (IFCB) automated cell imaging data, environmental sensor data, and predictive modeling to estimate *L. polyedra* abundance and communicate bloom dynamics through interactive visualization. The goal is to provide earlier, clearer signals of bloom risk that can support public awareness, coastal decision-making, and ecosystem protection.

1.3 Prior works

Harmful algal bloom forecasting has been approached using a variety of modern methods that address the nonlinear, multiscale nature of aquatic ecosystems.

A study by (Agarwal, Chávez-Casillas and Mouw 2023) applied empirical dynamic modeling (EDM) and automated cell imaging to make sub-monthly predictions of algal bloom abundance using Imaging FlowCytobot (IFCB) observations, demonstrating that multivariate models using image-derived features can improve prediction skill compared to univariate models.

(Özlem Baydaroğlu 2025) evaluated the performance of EDM for forecasting chlorophyll-a, a known indicator for algal blooms in Lake Erie. The authors showed that EDM can capture complex dynamics and produce accurate forecasts of chlorophyll-a concentrations despite irregular sampling and data sparsity from water measurements.

(Demiray et al. 2025) compared multiple deep learning models for HAB prediction using a suite of water quality, meteorological, and physical variables at Lake Erie to predict chlorophyll-a. Their results suggest neural networks can achieve strong predictive performance while providing insights into variable importance and model behavior.

(Szewczyk, Aleynik and Davidson 2025) demonstrated that ensemble forecasting methods can improve near-term forecasts of HAB occurrences associated with biotoxin risk.

Collectively, these studies establish that nonlinear time series methods outperform traditional linear approaches for bloom forecasting and that combining imaging data with environmental measurements improves predictive skill. However, several gaps remain. First, while EDM has been applied to generic phytoplankton abundance and chlorophyll-a as proxy indicators, few studies have directly targeted species-specific bloom forecasting for toxin-producing dinoflagellates like *Lingulodinium polyedra*, which exhibit distinct dynamics and pose specific public health risks. Second, prior work has not systematically compared EDM approaches against both classical time series baselines (e.g., ARIMA) and modern supervised learning methods (e.g., Random Forest) within a unified evaluation framework, making it difficult to assess the relative value of different modeling paradigms. Third, most forecasting studies focus exclusively on quantitative metrics without providing accessible visualization tools that translate model outputs into actionable insights for researchers and the public. These are the gaps that will be addressed within this project.

2 Description of Data

Data were collected using Imaging FlowCytobot (IFCB), an automated imaging instrument designed to image and quantify phytoplankton at high temporal resolutions (Sosik et al. 2003). IFCB works by continuously sampling seawater and capturing high-resolution images of individual plankton cells using laser-triggered imaging. (`https://mclanelabs.com/imaging-flowcytobot/`)

At IFCB Site 183, located at Scripps Pier, California (Latitude: 32.86671, Longitude: -117.25587), sampling seawater approximately every 20–25 minutes, with each sampling bin analyzing approximately 5 mL of seawater at a flow rate of 0.25 mL min^{-1} (15 mL $hour^{-1}$). Observations span from October 2022 through January 2026, covering approximately three years.

For each imaged cell in each bin, ranging from 0-2000+ cells, they were processed through automated image classification relying on extracted morphological and optical features with machine learning for cell identification (Olson and Sosik 2007). (`https://github.com/hsosik/ifcb-analysis`)

To provide environmental context for dynamics of blooms, high resolution environmental measurements were collected at Scripps Pier. These measurements were obtained from sensors, including conductivity-temperature-depth (CTD) instruments, SeaFET/SeapHox sensors, as well as barometers, and hygrometers. List of environmental variables that were collected and used are shown in (Table 1).

Table 1: Environmental Variables Associated Units

Variable	**Units**
Sea Water Temperature	°C
Salinity	PSU
Chlorophyll Concentration	μg L^{-1}
Conductivity	mS cm^{-1}
Sea Water Density	kg m^{-3}
Sea Water Pressure	dbar
Wind Speed	m s^{-1}
Wind Direction	degrees
Wind Gust	m s^{-1}
Air Temperature	°C
Barometric Pressure	mb
Water Level	m

For each sampling bin, header files (HDR) were used to extract sampling metadata, including flow rate, run duration, and ROI count. These parameters were required to derive mL analyzed and estimates of cell concentration (cells mL^{-1}).

Features file includes image-derived morphological and texture features extracted from each imaged cell, including measurements of cell size, shape descriptors, and color intensity. These features used provide quantitative descriptors of individual cells and are commonly used in plankton classification and bloom detection studies. Which includes a list of 240 features for each imaged cell (`https://github.com/hsosik/ifcb-analysis/wiki/Feature-File-Documentation`). 20 image features were selected as they show relevant morphological and ecological significance (Sonnet et al. 2022) (Agarwal, Chávez-Casillas and Mouw 2023). Image features shown in (Table 2).

Table 2: Image-Derived Features and Units

Feature	Units
Area	pixels^2
Biovolume	pixels^3
MajorAxisLength	pixels
MinorAxisLength	pixels
Perimeter	pixels
Orientation	degrees
Eccentricity	-
Solidity	-
texture_uniformity	-
texture_smoothness	-
texture_average_gray_level	gray-level intensity
texture_entropy	-
texture_average_contrast	-
H90	-
H180	-
Hflip	-
Extent	-
EquivDiameter	pixels
ConvexArea	pixels^2
ConvexPerimeter	pixels

Autoclass file includes automated classification outputs which assign probabilities for each detected particle representing the likelihood the image corresponds to a given taxon where each row corresponds to an imaged cell each column represents a taxa.

2.1 Data Collection

IFCB data was collected from Site 183 at Scripps Pier (`https://ifcb.caloos.org/bin?dataset=scripps-pier-ifcb-183`) using two approaches. Initially, a custom web-scraping pipeline was developed using `BeautifulSoup` and `Selenium` to automate data retrieval from the IFCB web dashboard. This script was run between January 5 and January 26, retrieving 120,000 files (115 GB of data) corresponding to 41,306 sampling bins.

After consultation with IFCB data maintainers, a more streamlined approach to data collection method was implemented using the IFCB Dashboard API. The API provides access to persistent identifiers (PIDs) for each sampling bin in json format, removing the need for browser-based scraping and improving speed and reliability of collection.

The same API based script was later used to collect data from the Del Mar Mooring IFCB site 158 during the week of February 2nd, 2026 for additional analysis.

Environmental data accompanying IFCB observations were obtained from the Southern California Coastal Ocean Observing System(SCCOOS) Data Portal and the NOAA Tides and

Currents station at Scripps Pier. These publicly available datasets provide high-resolution measurements of physical and chemical conditions concurrent with biological sampling.

2.2 Data Preparation

IFCB observations were aggregated and binned by day to reduce noise and to account for irregular sampling intervals and instrument downtime. For each bin, derived image features were summed using weighted averages based on the expected value of the taxas appearance. Where expected values were computed by summing probability of class.

To define the expected concentration of *L. polyedra* cells per milliliter, we followed IFCB concentration estimations provided in ifcb-analysis package to derive milliliters analyzed (`https://github.com/hsosik/ifcb-analysis`).

$$\text{ml analyzed} = (.25 \times (T_{\text{run}} - T_{\text{inhibit}})) \div 60$$

Environmental variables were aggregated and binned by day to align with IFCB derived abundance.

Missing values in both abundance and environmental time series were imputed using an exponential moving average (EMA) of 30 days. EMA is a smoothing method that estimates missing observations based on recent past values, assigning exponentially decreasing weights to older observations. This approach works well for short gaps in daily environmental and ecological time series, as it generally preserves temporal structure while avoiding artificial discontinuities and data leakage. Environmental variables with more than 30 days of missing data were excluded to prevent bias from excessive smoothing or extrapolation.

After imputation, all time series were normalized for comparability across features and ensure numerical stability in multivariate modeling.

$$x' = \frac{x - \mu}{\sigma}$$

where μ and σ denote the mean and standard deviation computed from the relevant analysis subset.

Three feature sets were created: image features only, environmental features only, and a combined set, containing both image features and environmental features.

3 Methods

3.1 Baseline Models

As benchmarks, we trained **Linear Regression**, **Ridge Regression**, and **Random Forest Regression** using IFCB image features with an 80/20 random train-test split. This baseline

helps assess how well morphology alone can predict bloom concentration, but it does not preserve temporal ordering.

For interpretability, we examined:

- the magnitude of Ridge coefficients after standardization, and
- impurity-based feature importance from Random Forest.

3.2 ARIMA Time Series

To establish a classical time series baseline that respects temporal ordering, we applied an autoregressive integrated moving average (ARIMA) model to the normalized L. polyedra concentration series. ARIMA models represent linear time series dynamics through autoregressive (AR) and moving average (MA) components, with optional differencing (I) to achieve stationarity (Box and Jenkins 1976).

We used automated model selection via the `pmdarima` package to identify the optimal ARIMA order, testing AR and MA orders up to 5. The selection procedure identified an ARIMA(4,0,0) specification based on information criteria, indicating that the series is stationary and follows a fourth-order auto-regressive process.

For multi-step forecasting, we implemented a rolling window approach: at each time point t (starting at the 80% mark of the series), the model was trained on all prior data and then forecasted T_p steps ahead. Performance was evaluated by comparing the forecast to the observed value at $t + T_p$, using Pearson correlation ρ, RMSE, and MAE across prediction horizons $T_p = 1$ to 31 days.

3.3 Empirical Dynamic Modeling

To better capture time dependence and nonlinear system behavior, we applied two EDM approaches.

Univariate Simplex Projection

For univariate predictions, we used simplex projections (Sugihara and May 1990) to forecast L. polyedra concentration using only the target series' own history. The target attractor was reconstructed from lagged copies of the target series with embedding dimension $E = 4$ and delay $\tau = 1$. For each prediction horizon T_p, we generated bootstrap forecasts using randomized library windows of 250 time steps.

Multiview Embedding (MVE)

To improve upon the univariate simplex, we applied Multiview Embedding (MVE) (Ye and Sugihara 2016), which reconstructs the attractor using multiple measured variables simultaneously rather than lagged copies of a single series. For each prediction horizon ($T_p = 1$ to 31 days), 500 randomized simplex projections were performed, each using three randomly selected predictors plus the target variable ($E = 4$). We compared three feature sets: (1) image features only (21 IFCB morphological variables including *Prorocentrum*

micans concentration), (2) environmental features only (12 oceanographic and meteorological variables), and (3) the combined set of all 33 features.

3.4 Evaluation Metrics

Model performance was evaluated using metrics suited to each modeling framework:

- Baseline supervised models: test R^2 and RMSE
- ARIMA: Pearson correlation ρ, RMSE, and MAE
- EDM models: Pearson correlation ρ, RMSE, and effective skill ρ_{eff}

For EDM models, predictive skill was evaluated using ρ_{eff}, defined as:

$$\rho_{\text{eff}} = \rho - |\text{ACF}(T_p)| \tag{1}$$

where ρ is the Pearson correlation between observations and predictions, and $|\text{ACF}(T_p)|$ is the absolute autocorrelation of the target series at lag T_p. This metric isolates genuine predictive skill beyond what a naive persistence forecast would achieve.

3.5 Visualization Implementation

To communicate model behavior over time, we built an interactive web visualization that maps daily observations into a multi-encoding bubble chart (temperature on x, pH on y, bloom magnitude by bubble size, chlorophyll by color), with playback controls and date scrubbing for temporal exploration. This interface supports qualitative validation of regime shifts, bloom clustering, and horizon-dependent predictability observed in the quantitative results.

4 Results

4.1 Baseline Models

As an initial benchmark, we trained standard supervised regression models to predict daily *L. polyedra* concentration (`Lpoly_expected_ml`), defined as the expected number of *L. polyedra* cells per milliliter of seawater as estimated by IFCB automated classification, from IFCB image-derived biological features using an 80/20 random train-test split. Sampling effort variables (e.g., volume analyzed, total image count) were excluded to ensure that model performance reflects genuine morphological signal rather than mechanical correlations with particle throughput.

The Random Forest model ($n_{\text{estimators}} = 80$, max_depth $= 10$) substantially outperformed both linear baselines, achieving a test R^2 of 0.523 and RMSE of 7.990. This suggests that the relationship between image-derived morphological features and *L. polyedra* concentration is meaningfully nonlinear.

Table 3: Baseline regression performance on a random 80/20 split using biological image features only.

Model	Test R^2	Test RMSE
Linear Regression	0.245	10.051
Ridge Regression	0.349	9.331
Random Forest	0.523	7.990

4.1.1 Feature Importance

We assessed feature importance using both Ridge coefficient magnitudes (after standardization) and Random Forest impurity-based importance (Table 4).

Table 4: Top 10 Random Forest feature importances (impurity-based), biological features only.

Feature	Importance
Ring10	0.208
moment_invariant1	0.201
moment_invariant7	0.094
Ring09	0.093
HOG41	0.083
moment_invariant2	0.081
HOG79	0.055
moment_invariant3	0.051
HOG81	0.040
Perimeter	0.039

The most important biological predictors fall into three categories: moment invariants (shape descriptors capturing rotational symmetry properties of cell outlines), ring features (radial intensity patterns measured at different distances from the cell centroid), and HOG features (histogram of oriented gradients summarizing local edge and texture structure). Ring10 and moment_invariant1 together account for over 40% of total impurity importance, indicating that radial intensity profiles and overall shape geometry are the strongest image-based indicators of *L. polyedra* concentration.

It is important to note that the Random Forest baseline uses a random train-test split, which does not respect the temporal ordering of the data. This means future observations may appear in the training set, potentially inflating performance through data leakage. The EDM-based models described below preserve temporal structure by design.

4.2 Univariate Simplex Projection

Table 5: Univariate simplex projection performance across selected prediction horizons. Each entry is the mean over 200 bootstrap trials with library windows of 250 time steps.

T_p (days)	ρ	ρ_{eff}
1	0.709	−0.042
2	0.751	0.244
3	**0.752**	**0.388**
4	0.497	0.173
7	0.250	0.088
10	0.131	−0.044
14	0.076	−0.035
20	0.129	0.108
25	0.157	0.153
30	0.121	0.121

The univariate simplex achieved its strongest effective skill at $T_p = 3$ ($\rho_{\text{eff}} = 0.388$), but produced negative ρ_{eff} at $T_p = 1$ and across much of the mid-range ($T_p = 8$ to 13), indicating that it fails to outperform the persistence baseline at these horizons. The negative ρ_{eff} at $T_p = 1$ is notable: while raw $\rho = 0.709$ appears strong, it is entirely attributable to the high autocorrelation at one-day lag. A modest recovery in skill appeared around $T_p = 20$ to 30 ($\rho_{\text{eff}} \approx 0.10$ to 0.15), hinting at longer-period structure that the univariate model captures only weakly.

4.3 Multiview Embedding

All three feature sets exhibited a characteristic two-peak structure in forecast skill (Table 6). The strongest predictive skill occurred at $T_p = 3$ days, where all models achieved $\rho_{\text{eff}} \approx 0.58$ and the combined model produced an RMSE of 0.387 on normalized data. Skill then declined sharply at $T_p = 4$ and remained low through $T_p \approx 13$, with RMSE approximately doubling to values near 1.0. A second peak in skill emerged around $T_p = 28$ to 30, where the combined model achieved ρ_{eff} of 0.452 at $T_p = 29$.

Table 6: MVE forecast skill (ρ_{eff}) and RMSE across selected prediction horizons for image, environmental, and combined feature sets. Each entry represents the mean over 500 randomized simplex projections.

T_p (days)	ρ_{eff} (Image)	ρ_{eff} (Env.)	ρ_{eff} (Combined)	RMSE (Combined)
1	0.189	0.180	0.186	0.445
2	0.442	0.434	0.439	0.409
3	**0.585**	**0.575**	**0.581**	**0.387**
4	0.270	0.262	0.262	0.817
8	0.079	0.218	0.102	0.990
13	0.020	0.092	0.099	1.005
19	0.359	0.363	0.379	0.946
25	0.439	0.257	0.353	0.970
29	**0.367**	**0.414**	**0.452**	**0.954**
31	0.311	0.353	0.397	0.956

The short-term peak at $T_p = 3$ likely reflects the characteristic timescale of bloom initiation and growth, during which the current state of the plankton community and its environment remain informative. The mid-range trough ($T_p = 4$ to 13) corresponds to horizons where short-term predictability is exhausted and the system exhibits its most irregular behavior. The second peak around $T_p = 28$ to 30 suggests a quasi-monthly periodicity in bloom dynamics, potentially driven by lunar tidal cycles or periodic upwelling events at Scripps Pier.

4.3.1 Feature Set Comparison

At short horizons ($T_p = 1$ to 3), all three feature sets performed comparably, indicating that the short-term dynamical signal is robust regardless of feature source. At intermediate horizons ($T_p = 6$ to 17), environmental features sustained higher skill than image features alone (e.g., $\rho_{\text{eff}} = 0.258$ vs. 0.142 at $T_p = 7$), suggesting that oceanographic conditions provide a longer predictive look-ahead than cell morphology. At longer horizons ($T_p = 28$ to 31), the combined feature set consistently outperformed both individual sets, achieving the highest ρ_{eff} at $T_p = 29$. This indicates that image and environmental variables provide complementary information for forecasting bloom dynamics beyond the short-term window.

4.3.2 MVE Feature Importance

Table 7: Top MVE predictors by proportion of appearance in the top 5% of models at $T_p = 1$. (CTD refers to Conductivity-Temperature-Depth sensor measurements.)

Image Features	Proportion
Pmicans_expected_ml	0.47
H90	0.29
Solidity	0.28
Hflip	0.22
Extent	0.21
Environmental Features	Proportion
sea_water_pressure_ctd	0.56
waterlevel_verified_m	0.43
sea_water_temperature_ctd	0.41
sea_water_electrical_conductivity_ctd	0.40
sea_water_practical_salinity_ctd	0.40

Feature importance was assessed by examining the frequency with which each variable appeared in the top 5% of models ranked by ρ_{eff} at each horizon (Table 7).

Among environmental predictors, sea water pressure (measured by CTD, Conductivity-Temperature-Depth sensor) was the dominant driver (proportion = 0.56 at $T_p = 1$), followed closely by water level (0.43), sea water temperature (0.41), and electrical conductivity (0.40). These environmental features maintained their dominance across forecast horizons, with pressure, water level, and temperature ranking among the top predictors throughout the 31-day forecast window.

Among image features, the concentration of *Prorocentrum micans*, another dinoflagellate species, was by far the most frequently selected predictor (proportion = 0.47 at $T_p = 1$), suggesting that blooms of these two species may be driven by similar environmental conditions or share ecological dynamics. Cell orientation metrics (H90, Hflip) and geometric properties (Solidity, Extent) were also important. Unlike environmental features, the relative importance of image predictors shifted substantially across horizons, suggesting that different aspects of cell morphology and co-occurring taxa become relevant at different forecast timescales.

4.4 ARIMA

Table 8: ARIMA(4,0,0) rolling forecast performance across selected prediction horizons.

T_p (days)	ρ	RMSE	MAE
1	0.319	0.063	0.063
3	0.194	0.152	0.152
7	0.180	0.216	0.216
14	0.174	0.261	0.261
25	0.206	0.275	0.275
31	0.232	0.277	0.276

The ARIMA model achieved its strongest correlation at $T_p = 1$ ($\rho = 0.319$) but exhibited relatively flat skill across longer horizons, with correlation remaining near 0.18 to 0.23 and RMSE gradually increasing from 0.063 at $T_p = 1$ to 0.277 at $T_p = 31$. Notably, the model shows no evidence of the two-peak structure observed in the EDM approaches. At short horizons ($T_p = 1$ to 3), ARIMA underperforms both the univariate simplex and MVE, suggesting that linear autoregressive structure alone is insufficient to capture the short-term nonlinear dynamics of bloom events. At longer horizons ($T_p > 20$), ARIMA maintains modest positive correlation but still falls short of MVE's second peak in skill.

The poor performance at $T_p = 3$, where MVE achieves $\rho_{\text{eff}} = 0.581$, is particularly revealing. ARIMA's $\rho = 0.194$ at this horizon is not adjusted for autocorrelation, meaning its effective skill would be even lower. This indicates that the persistence and linear AR structure captured by ARIMA cannot account for the rapid state changes characteristic of bloom initiation. The flat trajectory of ARIMA skill across horizons suggests that the model is primarily relying on slow drift and trend components rather than capturing the regime-switching behavior evident in the data.

4.5 Model Comparison

Table 9: Comparison of modeling approaches. Random Forest uses a random split (non-temporal), while ARIMA, Simplex, and MVE respect temporal ordering.

	Random Forest	ARIMA	Simplex	MVE (Combined)
Approach	Supervised	Linear TS	EDM (univariate)	EDM (multivariate)
Temporal structure	No (random)	Yes	Yes	Yes
Features used	Image	Target only	Target only	Image + Env.
Best metric	$R^2 = 0.523$	$\rho = 0.319$	$\rho_{\text{eff}} = 0.388$	$\rho_{\text{eff}} = 0.581$
Best horizon	N/A (static)	$T_p = 1$	$T_p = 3$	$T_p = 3$
Multi-step forecast	No	Yes	Yes	Yes
Captures nonlinearity	Yes	No	Yes	Yes

The Random Forest provides a strong static baseline ($R^2 = 0.523$) but does not produce temporal forecasts and its random split may inflate performance through data leakage. ARIMA respects temporal structure and produces multi-step forecasts, but its reliance on linear autoregressive dynamics limits its ability to capture bloom events, achieving only $\rho = 0.319$ at $T_p = 1$ and showing no skill peaks at biologically relevant timescales. The univariate simplex improves upon ARIMA by capturing nonlinear structure, achieving $\rho_{\text{eff}} = 0.388$ at $T_p = 3$, but is still limited to the target series' own history and frequently fails to beat persistence. MVE substantially improves upon all baselines: it provides genuine out-of-autocorrelation skill at short-term ($T_p = 3$, $\rho_{\text{eff}} = 0.581$) and monthly ($T_p = 29$, $\rho_{\text{eff}} = 0.452$) horizons, while combining image and environmental data yields the best long-range forecasts, supporting a multimodal monitoring approach for algal bloom prediction.

4.6 Interactive Data Visualization

[Link to Visualization]

To complement the quantitative modeling results, we developed an interactive bubble chart that visualizes 804 days of co-located sensor and plankton observations from Scripps Pier (October 2022 to December 2025). Each bubble represents a single day and simultaneously encodes four variables: sea water temperature (x-axis), sea water pH (y-axis), *L. polyedra* concentration (bubble size), and chlorophyll concentration (bubble color, on a green gradient from low to high). A 14-day trailing animation allows users to observe how environmental conditions evolve in the days leading up to and during bloom events. A bloom event timeline marks all days exceeding 5 mL, and users can click any event to jump directly to it.

The visualization reveals several patterns consistent with our modeling results. Bloom events (large bubbles) cluster within the 17–22°C temperature band, corroborating sea water temperature as the dominant environmental predictor identified by MVE feature importance (proportion = 0.76 at $T_p = 1$). Elevated pH values frequently co-occur with bloom conditions, reflecting the pH increase driven by photosynthetic carbon uptake during dense phytoplankton growth. High chlorophyll days (dark green bubbles) tend to precede or coincide with the largest bloom events, indicating that general phytoplankton productivity provides context for *L. polyedra*-specific blooms.

The temporal animation is particularly informative: by watching the 14-day trail, users can observe the characteristic trajectory of a bloom event, a gradual drift into the warm-temperature, elevated-pH region of the plot followed by a rapid increase in bubble size over 2 to 4 days. This visual timescale aligns with the $T_p = 3$ short-term predictability peak identified by both the simplex and MVE models. The animation also makes the seasonality of bloom events immediately apparent, with activity concentrated in late summer and fall.

This visualization serves as both an exploratory tool and a communication device. For researchers, it provides an intuitive way to identify candidate bloom drivers and verify that model-identified relationships (e.g., the temperature-bloom association) hold visually across the full observational record. For broader audiences, including coastal managers,

public health officials, and the general public, it translates complex multivariate sensor data into an accessible format that conveys when and under what conditions harmful blooms are most likely to occur, supporting awareness and preparedness for HAB events along the Southern California coast.

5 Discussion

5.1 Predictability of *L. polyedra*

Our results show that *L. polyedra* blooms can be predicted at two distinct timescales, with the strongest forecasts occurring 3 days ahead. The multiview embedding (MVE) model achieved an effective correlation of $\rho_{\text{eff}} = 0.581$ at 3 days, significantly outperforming both the univariate simplex ($\rho_{\text{eff}} = 0.388$) and ARIMA ($\rho = 0.319$). This 3-day window aligns with the typical bloom development timeline observed in similar dinoflagellate species.

Between days 4 and 13, forecast accuracy drops sharply. During this mid-range period, prediction errors roughly double and skill drops below 0.15 for all models, suggesting that the system becomes highly unpredictable at weekly timescales. This may be due to random environmental disruptions such as sudden wind events or changes in ocean currents that affect bloom development in ways that are difficult to anticipate.

Forecast skill recovers at 29 days ($\rho_{\text{eff}} = 0.452$ for combined features), revealing a monthly cycle in bloom patterns. This timescale matches lunar tidal cycles, which influence coastal water mixing, nutrient availability, and water movement patterns at Scripps Pier. Environmental features maintained better predictive power than image features at intermediate horizons (days 6 to 17), suggesting that ocean conditions, rather than the plankton population itself, drive the monthly bloom cycle. The combined model consistently outperformed individual feature sets at longer horizons, demonstrating that image and environmental data capture different but complementary aspects of bloom dynamics: image features reflect the current state of the plankton population, while environmental features indicate the conditions that will shape future bloom development.

ARIMA showed flat performance across all forecast horizons with no evidence of the two-peak pattern, highlighting the limitations of linear models for capturing bloom dynamics. The univariate simplex improved on ARIMA by accounting for nonlinear patterns but remained limited by using only the bloom concentration history itself. MVE's superior performance confirms that using multiple environmental and biological measurements together provides a more complete picture of the system than any single data source.

5.2 Feature Drivers of Bloom Dynamics

5.2.1 Environmental Predictors

Among environmental variables, sea water pressure was the most important predictor (proportion = 0.56), followed by water level (0.43), temperature (0.41), conductivity (0.40), and salinity (0.40). Pressure and water level likely serve as indicators of water column stability and mixing intensity, both critical for creating the calm, layered conditions that allow dinoflagellates to accumulate near the surface. Water pressure integrates multiple oceanographic processes including tides, waves, and water circulation patterns, making it a comprehensive indicator of suitable bloom conditions.

Temperature ranked third overall and remained among the top five predictors at all forecast horizons, confirming its established role as a primary bloom driver. The visualization supports this relationship, showing bloom events concentrated in the 17 to 22°C range. Conductivity and salinity, which respond to freshwater input and upwelling, likely signal nutrient delivery processes that support bloom initiation. The consistency of these environmental rankings across forecast horizons suggests that physical ocean conditions control bloom dynamics at both short-term (days) and monthly timescales.

5.2.2 Image-Derived Predictors

The concentration of *Prorocentrum micans*, another dinoflagellate species, was the most frequently selected image feature (proportion = 0.47), suggesting these two species respond to similar environmental conditions. Both are large dinoflagellates that prefer warm, stable, nutrient-rich waters, so high *P. micans* abundance may indicate favorable conditions for *L. polyedra*. Alternatively, both species may respond to the same ecological factors such as predation or competition.

Cell orientation and shape features (H90, Hflip, Solidity, Extent) were also important, potentially reflecting colony formation or stress responses that occur before blooms develop. Unlike environmental features, image feature importance varied across forecast horizons, indicating that different cell characteristics become relevant at different lead times. For example, cell size and shape may be most predictive during active growth (short horizons), while co-occurring species abundance may better indicate longer-term bloom persistence.

The Random Forest model, using only image features, achieved $R^2 = 0.523$ despite potential data leakage from its random split. While this confirms that cell measurements alone contain useful information, the fact that MVE combining image and environmental data outperformed image-only models demonstrates that multiple data sources are needed to fully capture bloom dynamics.

5.3 Study Limitations and Future Work

Several limitations affect the generalizability of these findings. First, all data come from a single location (Scripps Pier), meaning the model has learned patterns specific to this site's characteristics that may not apply elsewhere. Including data from Del Mar Mooring and other Southern California locations would test whether the model works across different coastal environments.

Second, using exponential moving averages to fill gaps in environmental data may smooth over rapid changes that are important for short-term forecasts. Future work should prioritize continuous sensor coverage or use more sophisticated gap-filling methods that preserve natural variability.

Third, the Random Forest model's random data split allowed future observations to influence training, potentially inflating performance. While this does not affect the time-respecting EDM models, it complicates comparisons. Retraining Random Forest with strictly chronological splits would provide a fairer benchmark.

Fourth, this analysis focused only on *L. polyedra*. Given that *P. micans* abundance was highly predictive, future studies should test whether the same approach can forecast multiple HAB species simultaneously. Expanding to include toxic species like *Pseudo-nitzschia* or *Alexandrium* would increase public health relevance.

Finally, these models currently work in retrospective mode using historical data. Deploying an operational early warning system would require real-time data feeds, automated model updates, and clear uncertainty estimates. Linking MVE forecasts with the interactive visualization could provide coastal managers and the public with 3-day and 29-day outlook products. Integration with existing HAB monitoring networks would enable broader adoption and validation against independent data.

Future research should investigate what causes the monthly bloom cycle at 29 days. While lunar tides are a likely driver, other processes such as internal waves, upwelling patterns, or water movement may also contribute. Examining how bloom timing relates to moon phases, combined with ocean circulation models, could identify the most important factors and improve long-range forecasts.

6 Conclusion

This study demonstrates that combining high-resolution imaging data from the Imaging FlowCytobot (IFCB) with environmental sensor measurements enables skillful short-term and monthly forecasting of *Lingulodinium polyedra* blooms at Scripps Pier. We evaluated four modeling approaches spanning classical time series methods (ARIMA), supervised machine learning (Random Forest), and nonlinear dynamical systems techniques (univariate simplex projection and multiview embedding). Multiview embedding (MVE) substantially outperformed all baselines, achieving effective correlation $\rho_{\text{eff}} = 0.581$ at a 3-day forecast horizon and $\rho_{\text{eff}} = 0.452$ at 29 days, demonstrating genuine predictive skill beyond what

naive persistence forecasts would achieve.

The analysis revealed a two-peak forecast skill structure that reflects distinct timescales in bloom dynamics: a 3-day window corresponding to bloom initiation and exponential growth, and a quasi-monthly periodicity likely driven by lunar tidal cycles or upwelling relaxation. Environmental features, particularly sea water pressure, water level, and temperature, dominated at all forecast horizons, while the concentration of co-occurring *Prorocentrum micans* emerged as the most informative image-derived predictor. The complementary nature of these data streams underscores the value of integrating multiple observational modalities for HAB forecasting.

The interactive visualization platform translates these quantitative findings into an accessible format, enabling coastal managers, public health officials, and the general public to explore bloom dynamics and understand the environmental conditions that promote harmful algal events. By revealing the clustering of bloom events within the 17 to 22°C temperature band and the characteristic 2 to 4 day bloom development trajectory, the visualization provides intuitive confirmation of model-identified relationships.

While this study establishes proof-of-concept for multimodal bloom forecasting at a single site, several avenues for extension remain. Expanding spatial coverage to include Del Mar Mooring and additional Southern California monitoring sites would test the model's ability to generalize and enable regional early warning systems. Transitioning from retrospective analysis to operational real-time forecasting would require automated data ingestion, model updating protocols, and uncertainty quantification frameworks. Extending the approach to additional HAB species, particularly toxin-producing taxa, would broaden public health relevance.

Ultimately, this work advances the goal of transforming HAB monitoring from reactive observation to proactive prediction. The demonstrated 3-day forecast skill provides a critical window for issuing beach advisories, adjusting shellfish harvest operations, and mobilizing sampling resources ahead of peak bloom conditions. The 29-day outlook, while more uncertain, offers strategic value for seasonal planning and resource allocation. By combining cutting-edge imaging technology, physically grounded dynamical modeling, and intuitive data visualization, this framework lays the foundation for next-generation HAB early warning systems that can mitigate the growing ecological and public health impacts of harmful algal blooms along the California coast and beyond.

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Appendices

A.1 Training Details A1
A.2 Additional Figures A3
A.3 Project Proposal A6

A.1 Training Details

A.1.1 Random Forest Configuration

The Random Forest regressor was trained using scikit-learn version 1.3.0 with the following hyperparameters:

- Number of estimators: 80
- Maximum tree depth: 10
- Splitting criterion: Mean squared error (MSE)
- Minimum samples per split: 2 (default)
- Minimum samples per leaf: 1 (default)
- Maximum features: Auto (sqrt of total features)
- Random state: 42 (for reproducibility)
- Number of jobs: 1 (single-threaded)

Training was performed on an 80/20 random split of the full dataset (N = 804 days). The training set contained 643 observations and the test set contained 161 observations. Feature standardization was applied via scikit-learn's StandardScaler before training for the Ridge regression model but not for Random Forest, as tree-based methods are invariant to monotonic transformations.

A.1.2 ARIMA Model Selection

ARIMA model selection was performed using the `pmdarima` package (version 2.0.3) with the following configuration:

- Search range: AR orders $p \in [0, 5]$, MA orders $q \in [0, 5]$
- Differencing: Automatically determined (d = 0 selected, indicating stationarity)
- Selection criterion: AIC (Akaike Information Criterion)
- Stepwise search: Enabled (for computational efficiency)
- Seasonal component: Disabled
- Error handling: Ignore warnings and continue search

The automated search identified ARIMA(4,0,0) as the optimal specification with the lowest AIC. Rolling forecasts were generated starting at 80% of the time series (observation 643) to ensure sufficient burn-in period. For each forecast horizon T_p, the model was retrained on all data up to time t and forecasted T_p steps ahead, then evaluated against the held-out observation at $t + T_p$.

A.1.3 Simplex Projection Bootstrap

Univariate simplex projection used the pyEDM library (version 1.14.0) with the following parameters:

- Embedding dimension: $E = 4$
- Time delay: $\tau = 1$
- Library window size: 250 time steps
- Number of bootstrap trials: 200
- Prediction range: Full series (1 to N)
- Nearest neighbors: $E + 1 = 5$ (default for simplex)

For each bootstrap trial at each prediction horizon T_p, a random starting point was selected uniformly from the valid range $[1, N - 250]$, and a consecutive 250-step window was used as the library set. The simplex algorithm identified the $E+1$ nearest neighbors in the reconstructed attractor space (using Euclidean distance) and weighted their future trajectories to generate predictions. Performance metrics (Pearson correlation ρ, RMSE, MAE) were averaged across all 200 bootstrap trials to produce stable estimates of forecast skill.

A.1.4 Multiview Embedding Configuration

MVE used the same pyEDM simplex core with the following modifications:

- Embedding dimension: $E = 4$ (target + 3 randomly selected predictors)
- Time delay: $\tau = 1$ (concurrent measurements, not lagged)
- Library: Full time series (1 to N)
- Prediction range: Full series (1 to N)
- Number of randomized trials: 500 per horizon
- Feature pool: 21 image features (morphological + *P. micans*), 12 environmental features, or 33 combined

For each of the 500 trials at each horizon T_p, three predictors were randomly sampled without replacement from the feature pool. These three predictors plus the target variable formed the 4-dimensional embedding space for that trial. All time series (target and predictors) were z-score normalized prior to embedding. Effective correlation ρ_{eff} was computed by subtracting the absolute autocorrelation $|\text{ACF}(T_p)|$ from the raw Pearson correlation ρ

for each trial, then averaged across all 500 trials.

Feature importance was assessed by ranking all 500 models by ρ_{eff} and examining the top 5% (25 models). For each feature, its importance was defined as the proportion of top models in which it appeared (i.e., frequency of selection divided by 25).

A.1.5 Computational Environment

All models were trained and evaluated on a Windows 11 LTSC system with the following specifications:

- Python version: 3.11
- CPU: 8 core processor
- RAM: 16 GB
- Key packages: scikit-learn, pyEDM, pmdarima, pandas, numpy

Total computational time:

- Random Forest training: ∼15 seconds
- ARIMA rolling forecasts (31 horizons): ∼3 minutes
- Simplex bootstrap (200 trials × 30 horizons): ∼8 minutes
- MVE (500 trials × 31 horizons × 3 feature sets): ∼15 minutes

Random seed 42 was used throughout to ensure reproducibility of all stochastic components (Random Forest bootstrap sampling, ARIMA optimization, random feature selection in MVE).

A.2 Additional Figures

This section presents additional visualizations supporting the model results and analysis discussed in the main text.

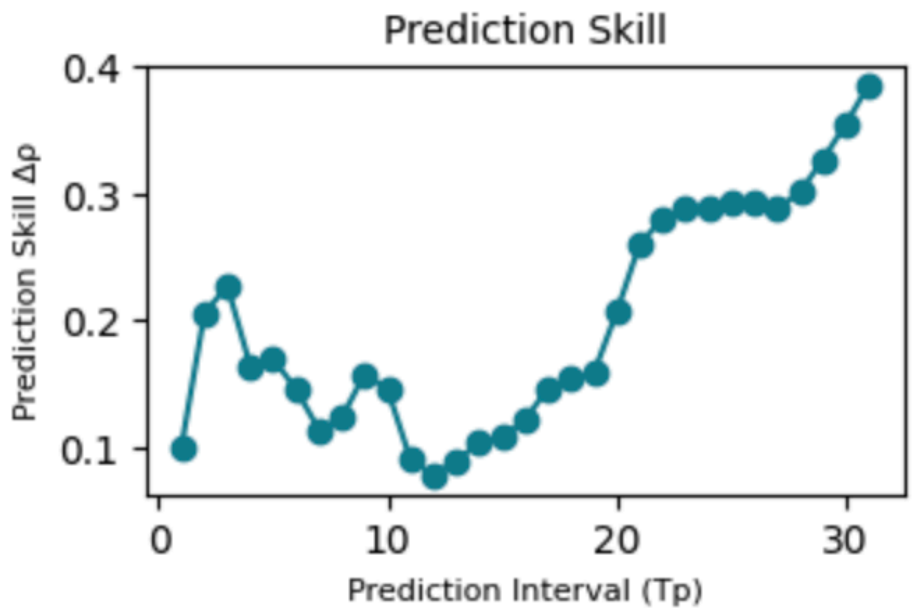

Figure A 1: Effective correlation (ρ_{eff}) as a function of prediction horizon (T_p) for the Multi-view Embedding model. The plot shows two distinct peaks at $T_p = 3$ days and $T_p = 29$ days, indicating short-term and monthly predictability windows for *L. polyedra* bloom dynamics.

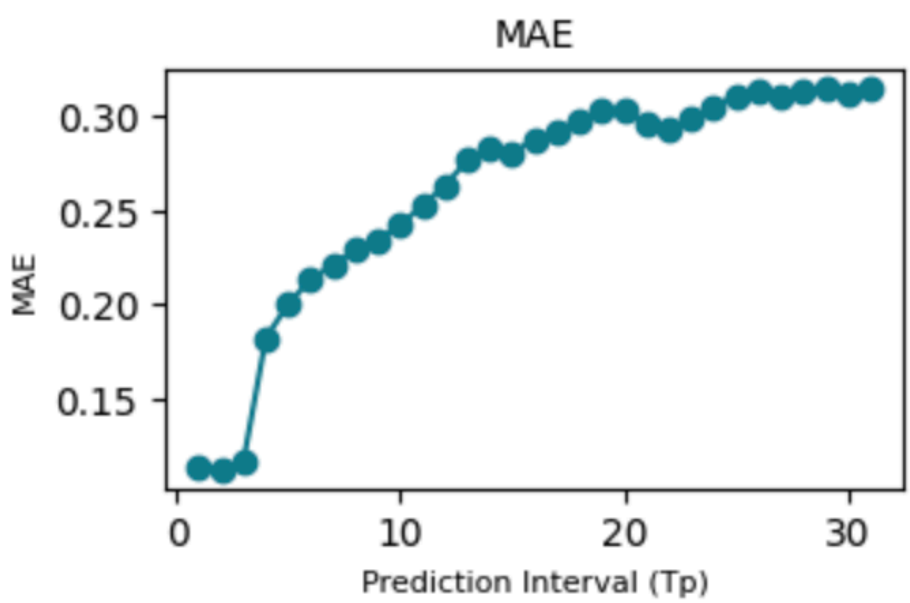

Figure A 2: Mean Absolute Error (MAE) as a function of prediction horizon (T_p) for the MVE model. Lower MAE values at $T_p = 3$ and $T_p = 29$ days correspond to the two predictability windows observed in the ρ_{eff} profile.

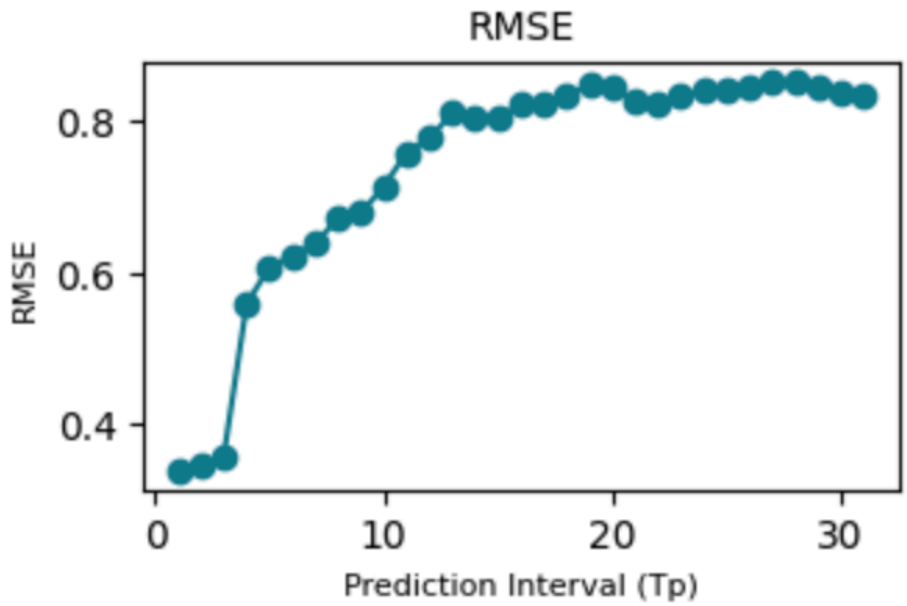

Figure A 3: Root Mean Squared Error (RMSE) as a function of prediction horizon (T_p) for the MVE model. The pattern closely mirrors the MAE profile, with prediction errors minimized at the same forecast horizons.

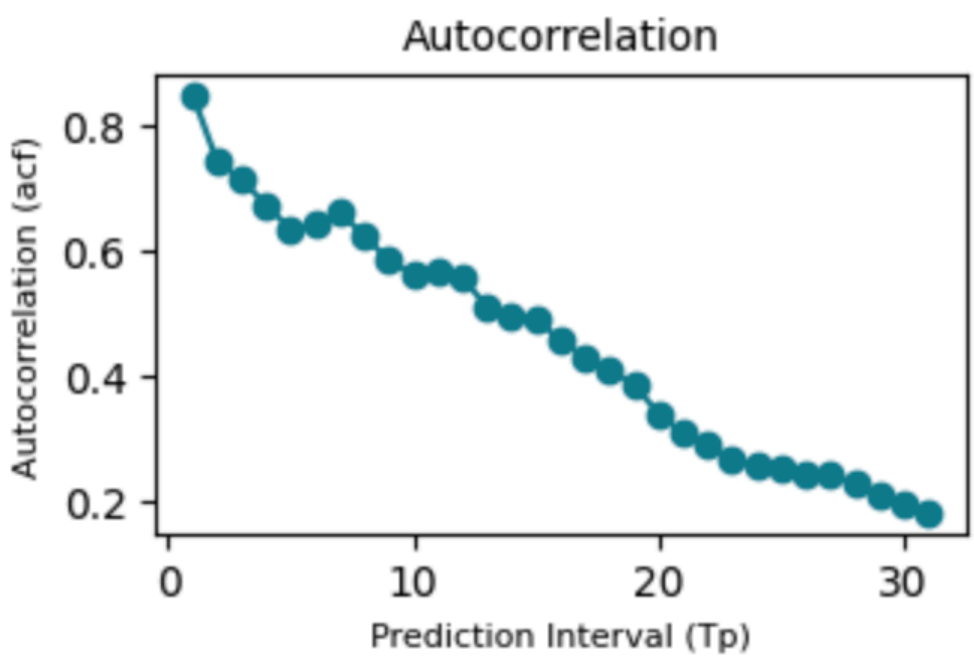

Figure A 4: Autocorrelation function (ACF) of the target time series as a function of lag. This baseline autocorrelation is used to compute ρ_{eff}, which adjusts model performance for the natural persistence in bloom concentration over time.

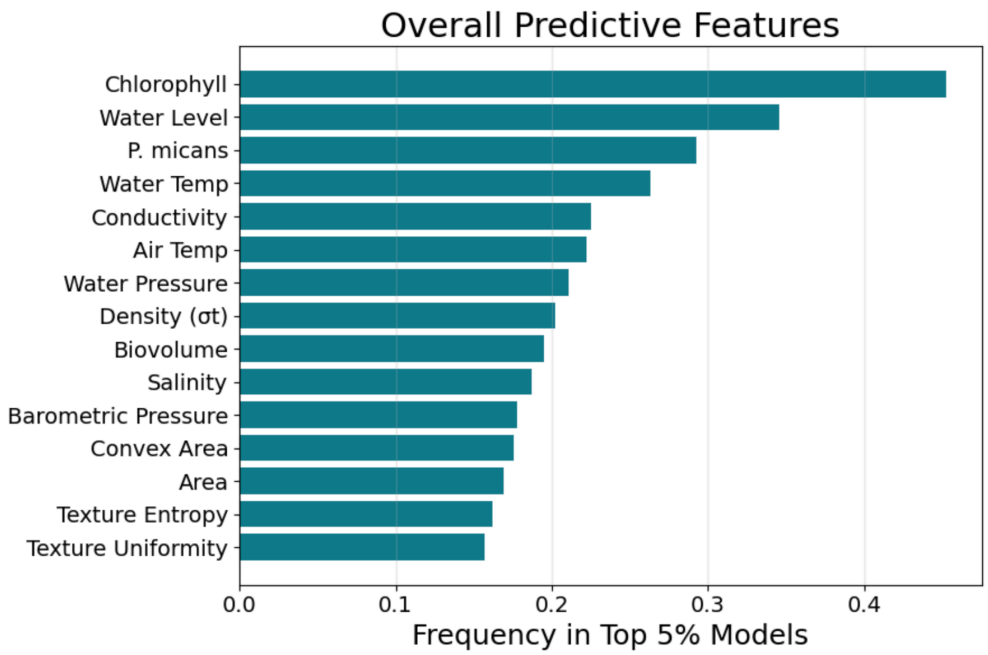

Figure A 5: Feature importance scores from the Random Forest model baseline. The top three features are Ring10, moment_invariant1, and moment_invariant6, all derived from IFCB morphological image analysis of plankton cells.

A.3 Project Proposal

For reference, the project proposal can be found here.